

Algorithms for genomic data analysis

Mapping high error sequencing reads

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Approximate matching for low similarity strings

- ▶ exact string match indexes can be used to approximate string matching as long as the similarity of the strings is high
- ▶ alternative approaches may perform better with 3rd generation sequencing technologies due to their characteristics:
 - ▶ high error rates ($\geq 10\%$)
 - ▶ long reads ($\geq 1kbp$)
- ▶ recall the idea of the partitioning approach:
 1. filter potential locations with text index for exact matching
 2. verify candidate locations
- ▶ areas of low similarity to the overall read should be identified rather than very similar short fragments

Sequence alignment and k -mer content similarity

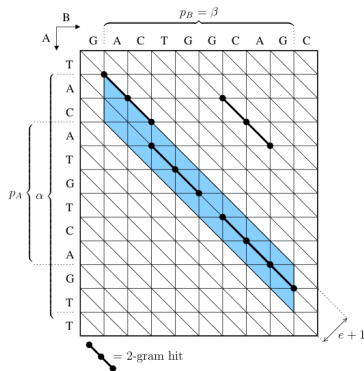


Fig. 1. An 8×2 -parallelgram in the edit graph between the sequences $A = \text{TACATGTCAGTT}$ and $B = \text{GACTGGCAGC}$. For $q = 2$, there are four q -hits within the parallelgram and $\text{dist}_\delta(\alpha, \beta) = 2$.

Lemma 1. Let α and β be substrings of A and B , respectively, s.t. $|\beta| = n$ and $\text{dist}_\delta(\alpha, \beta) \leq e$. Then, there exists an $n \times e$ parallelgram such that (a) it contains at least $T(n, q, e) := (n + 1) - q(e + 1)$ q -hits, (b) its B -projection is β , and (c) its A -projection is contained in α .

How to measure similarity between k -mer content?

Hypergeometric distribution describes the probability of m successes (random draws for which the object drawn has a specified feature) in n draws, without replacement, from a finite population of size N that contains exactly M objects with that feature.

In the case of k -mer content of DNA sequences:

- ▶ $N = 4^k$ (the universum of all possible k -mers)
- ▶ $M = |A|$
- ▶ $n = |B|$
- ▶ $m = |A \cap B|$

$|A \cap B|$ follows the hypergeometric distribution under the assumption that sequences A and B are independent.

Jaccard index is a "normalized" version of the above statistics

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Jaccard index vs sequence similarity

Jaccard index vs error rate

$$\mathcal{G}(\epsilon, k) = \frac{1}{2e^{\epsilon k} - 1} \quad \text{and} \quad \mathcal{F}(J, k) = \frac{-1}{k} \times \log\left(\frac{2J}{1+J}\right), \quad (1)$$

where $\mathcal{G}(\epsilon, k)$ serves as an estimate of the Jaccard similarity given an error rate and $\mathcal{F}(J, k)$ estimates the converse. $\mathcal{F}(J, k)$ can be shown as the maximum likelihood estimator (MLE) for error rate (proof in Section 10). Note $\mathbb{E}(J) \geq \mathcal{G}(\epsilon, k)$ (using Jensen's inequality).

Application to read mapping

Given a read A and the maximum per-base error rate $\epsilon_{\max}$, our goal is to identify target positions in reference B , where A aligns with $\leq \epsilon_{\max}$ per-base error rate. This problem can be exactly solved in $O(|A| \cdot |B|)$ time by designing a suitable quadratic time alignment algorithm. When mapping to a large database of reference sequences, solving this problem exactly is computationally prohibitive. Hence, we define an approximate version of this problem using the Jaccard coefficient as a proxy for the alignment as follows: Let B_i denote the substring of size $|A|$ in B starting at position i ($0 \leq i \leq |B| - |A|$). For a given k , we seek all mapping positions i in B such that

$$J(A, B_i) \geq \mathcal{G}(\epsilon_{\max}, k) - \delta \quad (4)$$

Note that if A aligns with B_i with per-base error rate $\leq \epsilon_{\max}$, then $\mathbb{E}(J(A, B_i)) \geq \mathcal{G}(\epsilon_{\max}, k)$ (using Eq. 1). As this equation applies only to the expected value of $J(A, B_i)$, we lower this threshold by δ to account for variation in the estimate. The parameter δ is defined as the margin of error in Jaccard estimation using a 90% confidence interval.

How to use k -mer content in practice?

Problems

- ▶ explicitly storing k -mer content of sequence A require $(|A| - k + 1) \cdot k$ bytes – much more than $|A|$
- ▶ calculating union/intersection of large sets may be slow
- ▶ it's even more difficult when storing the k -mer content for each potential pattern location in the text

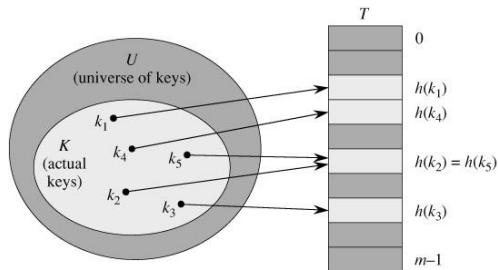
What we need?

- ▶ reduced representation of
 1. particular k -mers
 2. k -mer sets
- ▶ reduced representation of k -mer sets should enable estimation of Jaccard index

Reduced representation of particular k -mers

Solution:

Apply hash function.



- Collisions add noise, but its level may be controlled by adjusting the range of the hash function values

Reduced representation of k -mer sets

Observation

Let A and B be two finite sets. Consider randomly chosen linear ordering $\leq$ on $A \cup B$. Then

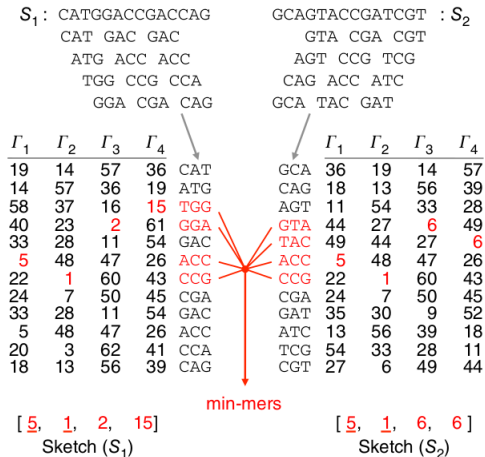
$$P(\min_{\leq}(A) = \min_{\leq}(B)) = J(A, B)$$

Solution:

Create a *sketch* of a k -mer set by choosing its *minhashes*: elements with lowest values of hash functions.

Minhash sketches of sequences

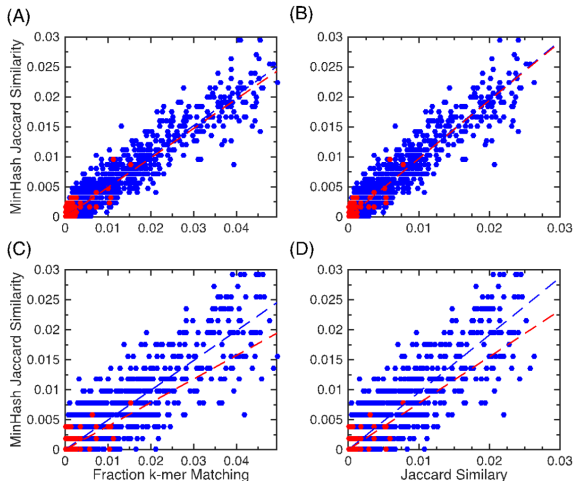
- sketch is the vector of the smallest values according to s independent hash functions
- $J(A, B)$ is estimated by the proportion of coordinates with equal values in $S(A)$ and $S(B)$



$$J(S_1, S_2) \approx 2/4 = 0.5$$

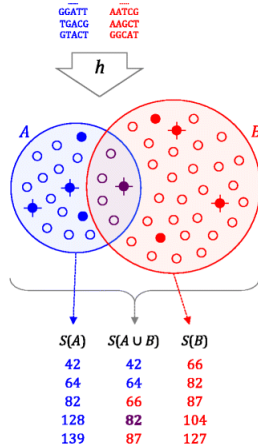
Jaccard index estimation quality

- ▶ each dot represents a pair of 10kbp long reads
- ▶ colors:
 - no overlap
 - overlap 20%
- ▶ sketch size:
 - 1256 (up)
 - 512 (down)



Sketches using single hash function

- ▶ sketch consists of s smallest values according to a single hash function
- ▶ unbiased JI estimation require calculating $S(A \cup B)$

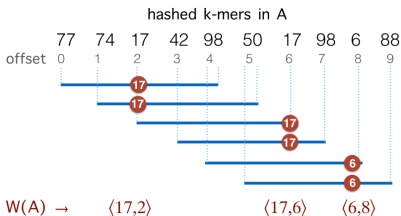


$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \approx \frac{|S(A \cup B) \cap S(A) \cap S(B)|}{|S(A \cup B)|}$$

Winnowing scheme

How to ensure that the k -mers selected for the sketch are evenly distributed across the sequence?

- ▶ $W(A)$ contains *minimizers*: minhashes sampled from consecutive windows of fixed size



Winnowing scheme – properties

Observation

Let $W(A)$ and $W(B)$ denote minimizer sets obtained using a winnowing scheme for window size w . Then:

- 1. every common subsequence of A and B of length $\geq w + k - 1$ is indicated by a common minimizer of $W(A)$ and $W(B)$,*
- 2. common subsequences of A and B of length $< k$ are ignored.*

Minimizer density

- ▶ *density* of a k -mer selection scheme W for a sequence A is defined by $d(W, A) = \frac{|W(A)|}{|A|-k+1}$
- ▶ *density factor* of a winnowing scheme W for a sequence A is defined by $d_f(W, A) = d(W, A) \cdot (w + 1)$

Observation

$$1 \leq d_f(W, A) \leq w + 1$$

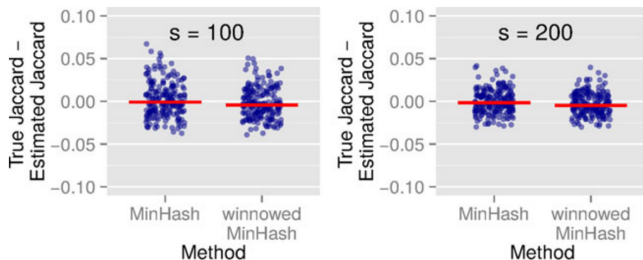
k-mer ordering comparison for winnowing scheme

	Ordering	d_f	distance mean $\pm$ stdev	low sep. %
dbg	lexico.	2.18	5.5(34)	27
	random	2.00	6.0(32)	18
	Minimap	2.05	5.9(32)	21
	KMC2	1.97	6.1(32)	18
	UMD Ovl	1.91	6.3(30)	14
	Kraken	1.88	6.4(29)	11
	DOCKS	1.75	6.9(25)	4.6
human	lexico.	2.34	5.1(34)	33
	random	2.02	6.0(32)	19
	Minimap	2.09	5.8(33)	22
	KMC2	2.02	5.9(33)	19
	UMD Ovl	1.97	6.1(31)	17
	Kraken	1.93	6.2(30)	13
	DOCKS	1.77	6.7(26)	6.2

Joining winnowing scheme with sketches

define *winnowed-minhash* estimate $\mathcal{J}(A, B_i)$ for $J(A, B_i)$ as

$$\mathcal{J}(A, B_i) = \frac{|S(W(A) \cup W(B_i)) \cap S(W(A)) \cap S(W(B_i))|}{|S(W(A) \cup W(B_i))|}$$



mean error by our method in estimating Jaccard similarity is <0.003